

Punjab National Bank

Two clocks, one engine — planned and reactive next-best-action from a single decisioning core

A day plan computed overnight and a nudge fired in seconds look like two systems. They are two clocks on one engine — and building them separately is how banks end up with two versions of what to say to the same customer.

2

clocks, one
decisioning core

1

arbiter for
both

3

hard gates that
remove actions

1

override that
beats margin

0

duplicate
ranking logic

1

closed loop
feeding both

Two scope items that read as separate builds are one decisioning core on two clocks — and keeping them together is what stops the bank saying two things at once

Situation

A large bank already owns its relationship platform. Two capabilities were prioritised: customer profile with offer management and next-best-action, and intelligent insights and nudges for cross-sell and up-sell.

Read separately, the first is a planning capability and the second is an alerting one.

Complication

Built separately, they each need customer context, eligibility rules, propensity scoring, value estimation, channel fitting and fatigue management. So both get built, twice.

Worse, they will disagree. A day plan says call this customer about a deposit; a nudge two hours later says offer an overdraft. Both are individually reasonable and together they are noise, because nothing arbitrated between them.

The architecture

One decisioning core, two clocks. A planned clock scores the portfolio overnight and delivers a ranked day plan by the time the relationship manager logs in. A reactive clock listens for events and fires a nudge within seconds while the situation is still live.

Same candidate generation, same scoring, same arbiter, same gates, same fatigue rules. Only the trigger and the latency differ.

THE NUMBERS THAT MATTER

1

engine, two triggers

1

arbiter, one ranking

3

hard gates

1

churn override

2

latency profiles

1

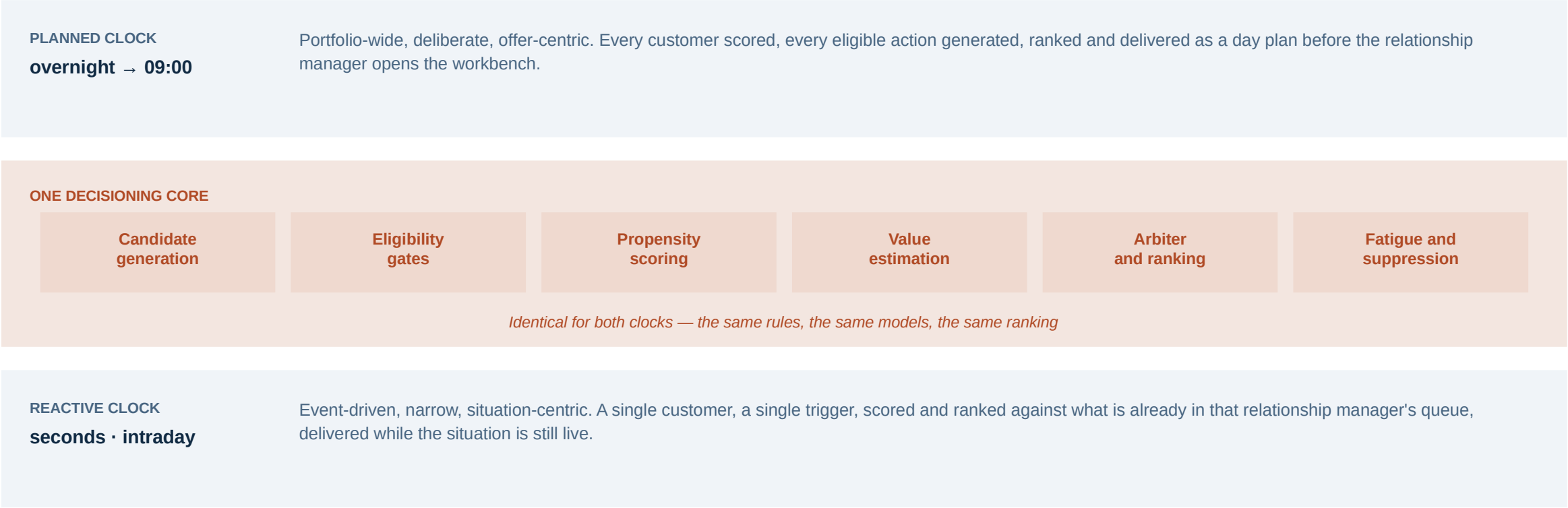
feedback loop

The positioning that matters commercially: this is not a replacement for the platform the bank already owns. It is the intelligence and orchestration layer above it — nothing is ripped out, and the value comes from acting on data the bank already has.

The scope items and platform position are the client's; the two-clock architecture, arbiter design and gating model are mine

SCOPE ITEMS	ARCHITECTURE ANALYSIS	ILLUSTRATIVE	OUT OF SCOPE
Client context The two prioritised capabilities and the existing platform position, as stated by the bank. • Profile and offer management • Insights and nudges for cross-sell • CRM platform already owned • Relationship-manager population • Existing systems of record • Additive layer expectation	My contribution The single-engine argument, the two-clock design, the arbiter and gating model and the feedback loop. • Two-clock architecture • Shared arbiter and ranking design • Hard-gate versus weighting position • Churn override design • Closed-loop feedback model • Executive briefing authored	Modelled Worked examples, latencies and volumes illustrate the design. Directional, not measured. • Worked day-plan example • Event-to-nudge latency profile • Candidate and suppression volumes • Replace with bank actuals • Shape is the argument • Method transfers unchanged	Not claimed Excluded deliberately. A pre-demonstration architecture position, not a delivery record. • Delivery of this capability • Any customer or portfolio data • The bank's models or thresholds • Any commercial position • Whether the pursuit proceeded • Benchmark figures as committed levels

One core, two triggers — the difference between the clocks is latency and cadence, not logic



Why one core rather than two systems

Both clocks need the same six stages. Built separately, each maintains its own eligibility rules, its own propensity models and its own ranking — and they drift apart within months, because nobody synchronises two rule sets that nobody has to reconcile.

The visible symptom is a relationship manager receiving a nudge that contradicts the day plan, at which point both are ignored.

HOW IT RUNS

The same engine, the same customer, two different moments — and the second one is only useful because it arrives in seconds

PLANNED — OVERNIGHT TO NINE	
Overnight	A deposit maturing in three weeks is detected across the portfolio scan
Scored	Retention propensity, reinvestment value and channel fit computed
Gated	A higher-margin product is removed entirely on suitability, not down-weighted
Ranked	Placed at the top of the relationship manager's day plan
09:00	Delivered as a prioritised action with the context needed to make the call

REACTIVE — SECONDS, INTRADAY	
10:40	A large credit lands in the same customer's current account
Event	The balance movement fires against a listening rule, not a nightly scan
Scored	Same models, one customer, evaluated against the live queue
Gated	Same eligibility, suitability and consent checks apply unchanged
10:40 + 8s	An idle-funds nudge reaches the relationship manager while the money is still in play

Why the reactive clock cannot simply be a faster batch

The value of the idle-funds nudge decays over hours, not days. A customer who has just received a large credit will decide what to do with it that morning; a nudge in tomorrow's day plan arrives after the decision.

That is the whole reason both clocks exist. The planned clock covers the portfolio comprehensively and cannot be fast. The reactive clock covers one customer at a time and cannot be comprehensive. Neither substitutes for the other, and both need to agree — which is why they share an engine rather than sharing a roadmap.

Hard gates remove actions entirely — they are never out-weighed by margin, and retention outranks the sale

01

Hard gates, not weights

- Eligibility, suitability and consent can remove an action from the candidate set completely.
- They are not inputs to a score. A weighted model can always be outvoted by a high enough margin — a gate cannot.
- This is the difference between a system that occasionally recommends something inappropriate and one that structurally cannot.

02

Retention outranks the sale

- Above a churn-risk threshold, calling the at-risk customer outranks any cross-sell for that customer.
- Implemented as an override on the ranking, not as a weight inside it.
- Without it, the highest-value action for a customer about to leave is almost always a sale — which is exactly backwards.

03

Then rank on value

- What survives the gates is ranked on propensity, value, margin and channel fit.
- Fatigue decay reduces the rank of anything this customer has recently been approached about.
- Capacity is respected — a day plan longer than a working day is a list, not a plan.

The closed loop

- Every outcome — accepted, declined, not attempted, and the reason — returns to the engine.
- That retrains propensity and recalibrates fatigue, and it is the only reason the ranking improves rather than ossifying.
- It also produces the evidence for a return-on-investment case measured on the bank's own pilot data rather than on a benchmark.

What has to be confirmed before positioning

- The integration mechanism to source systems, which determines what the reactive clock can actually listen to.
- The licensed-module boundary on the existing platform, which determines what is genuinely additive.
- That benchmark figures are presented as benchmarks and never as committed service levels.

Why gates are a governance artefact

- A gate is auditable in a way a weight is not — you can show that an action was never a candidate.
- With weights, the answer to why was this offered is a model explanation rather than a rule.
- In a regulated distribution context, that difference is the whole conversation with a supervisor.

Four rules for decisioning that has to run on more than one clock

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|----|---|---|
| 01 | One engine, several triggers | Planned and reactive decisioning need identical candidate generation, gating, scoring and ranking. Building them as separate systems duplicates all of it and guarantees they will disagree within months. |
| 02 | Gate, do not weight, on suitability | Eligibility, suitability and consent must remove an action from the candidate set. A weighted model can always be outvoted by a large enough margin, and cannot show a supervisor that an action was never available. |
| 03 | Let retention override value | Above a churn threshold, calling the at-risk customer beats any cross-sell. Implemented as an override rather than a weight, because a weight will lose to a sufficiently valuable sale exactly when it should not. |
| 04 | Close the loop or the ranking ossifies | Outcomes and reasons must return to the engine. Without that, propensity never improves, fatigue never calibrates, and the business case rests on benchmarks instead of the bank's own data. |

Two prioritised scope items looked like two systems. They were one decisioning core on two clocks — and recognising that early is the difference between an intelligence layer and two competing recommendation engines.